

Guideline

Potassium Supplementation - Intravenous

Reference #: HRM014

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	All areas	Medical, Nursing, Pharmacy

HIGH RISK MEDICATION

A high-risk medication alert exists for the use of intravenous potassium.
Hyperkalaemia can develop rapidly, remain asymptomatic and be potentially fatal.
Inappropriate administration of undiluted, intravenous potassium has been implicated in multiple fatalities.

Please use with caution and comply with recommendations within this document.

Infusion rates >10mmol/hr are not approved at Osborne Park Hospital.
Patients requiring these faster infusion rates must be transferred to SCGH.

Introduction¹⁻³

Intravenous potassium is used for potassium replacement therapy in severe hypokalaemia, or in mild / moderate hypokalaemia ONLY when oral supplementation is not possible. The risks associated with the use of intravenous potassium are well known. Errors relating to the preparation and administration of intravenous potassium can be and have been fatal.

The most common type of errors relating to intravenous potassium use involve:

- **Rapid administration:** Deaths have occurred from rapid administration of potassium.
- **Selection of the wrong ampoule:** Potassium chloride ampoules have been mistaken for sodium chloride 0.9% ampoules when reconstituting a drug.
- **Cognitive error:** Potassium chloride has been administered instead of furosemide (likely due to familial pairing of potassium supplementation with furosemide).
- **Preparation error:** Incorrect preparation of the intravenous potassium solution.
- **Inadequate inversion (mixing) of final solutions:** Following addition of concentrated potassium ampoules to a bag without adequate mixing.

A common root cause of potassium related errors is the availability of concentrated potassium chloride ampoules as stock on general ward areas.

Due to potential risks associated with using IV potassium ampoules, its availability is restricted and there are strong recommendations to use pre-mixed potassium solutions preferentially in ALL patient care areas (including critical care).

INTRAVENOUS POTASSIUM SUPPLEMENTATION – ADVICE SUMMATION

PERIPHERAL VENOUS ACCESS		CENTRAL VENOUS ACCESS	
RATE OF ADMINISTRATION	Maximum rate of 10mmol per hour Exception: when prescribed as per Diabetic Ketoacidosis (DKA) / Hyperosmolar Hyperglycaemic State (HHS) guidelines allowing up to 20mmol per hour via peripheral access Infusion rates >10mmol/hr are not approved at Osborne Park Hospital. Patients requiring these faster infusion rates must be transferred to SCGH.	RATE OF ADMINISTRATION	Maximum rate of 20mmol per hour Rates exceeding 20mmol per hour only under direction or advice of a Critical Care Specialist or other Consultant <i>If potassium-phosphate salt is administered the maximum rate allowed is 10mmol per hour (due to phosphate content) unless in a critical care area</i>
CONTINUOUS ECG MONITORING	Not required for rates ≤ 10mmol per hour Exception: when rate exceeds 10mmol per hour when prescribed as per DKA / HHS guidelines – continuous cardiac monitoring is required	CONTINUOUS ECG MONITORING	Required if rate exceeds 10mmol per hour
EXAMPLES OF ACCESS	Intravenous cannula (IVC)	EXAMPLES OF ACCESS	Central venous catheter (CVC), Peripherally inserted central catheter (PICC)

STANDARD PREMIXED POTASSIUM BAGS	- First line option for intravenous supplementation of potassium - Sterile, non-pyrogenic solutions – DO NOT ADD TO THESE BAGS - Reduces risk of phlebitis and potential clinical errors
NON-STANDARD POTASSIUM SOLUTIONS	- Acceptable in critical care settings under supervision of critical care staff (medical/nursing) - May be considered in exceptional circumstances where standard premixed solutions are deemed unsuitable (requires senior medical staff member approval) - Consider central access for administration of non-standard, hypertonic solutions
NEAT (undiluted) POTASSIUM SOLUTIONS	Permitted for use <u>strictly</u> in critical care areas following approval by the Consultant/SR and under close supervision by experienced nursing staff only
BOLUS ADMINISTRATION of POTASSIUM	- A bolus dose of 5mmol of potassium may be administered by intravenous injection in resuscitation emergencies. - NEVER administer potassium subcutaneously or intramuscularly

INTRAVENOUS POTASSIUM SUPPLEMENTATION – RECOMMENDATIONS

The following dosage recommendations are intended to provide guidance for intravenous potassium supplementation for a standard patient of average size (e.g. 70kg) with normal renal function and acid-base status.

Please use this table as a **general guide only** to assist with any clinical decisions relating to potassium supplementation in your patient.

Deviation from this guide may be required to appropriately supplement your patient, with consideration of any patient factors influencing potassium loss, gain or maintenance.

For management of hypokalaemia in DKA/HHS please refer directly to the respective clinical guidelines available for guidance.

INTRAVENOUS POTASSIUM SUPPLEMENTATION GUIDE

Clinical scenario	POTASSIUM CONCENTRATION		
	Mild Severity Hypokalaemia (3.0 to 3.4mmol/L)	Moderate Severity Hypokalaemia (2.5 to 2.9mmol/L)	Severe Hypokalaemia (< 2.5mmol/L)
Maintenance therapy	If tolerating and absorbing oral therapy and asymptomatic from perspective of hypokalaemia, then oral potassium supplementation may be considered. - Potassium chloride SR 600mg tabs (2 PO TDS) If intravenous supplementation necessary 60-80 mmol IV potassium over 24 hr period in divided doses	Intravenous supplementation 80-100 mmol IV potassium over 24 hr period in divided doses	Intravenous supplementation 100-120 mmol IV potassium over 24 hr period in divided doses OR alternatively 5 to 10mmol IV potassium per hour with regular serum potassium level monitoring to assess for ongoing requirements
As a general estimate, serum potassium levels are expected to increase by approximately 0.25mmol/L following IV administration of 20mmol potassium			
Other potential contributing factors to consider	DKA/HHS (refer to respective clinical guidelines) Nausea / vomiting Gastrointestinal infection Concurrent medications (e.g., diuretics, corticosteroids, insulin, beta-2 agonists, amphotericin B, laxatives) Renal impairment / dialysis TPN Malnutrition / refeeding syndrome / magnesium deficiency Ileostomy Endocrine conditions (Cushing's syndrome, Conn's syndrome, hyperaldosteronism) Nephrogenic diabetes insipidus		

Definitions

Standard 'pre-mixed' potassium solution	Sterile, proprietary intravenous preparations containing a known quantity of potassium, diluted in a compatible fluid. Not intended for any manipulation or addition of other medication/ electrolyte into these preparations.
Non-standard potassium solution	A non-proprietary intravenous solution containing an amount of potassium as prescribed on a medical order, that is aseptically prepared by a health professional.
Central administration	In reference to the delivery of a medicine by the intravenous route through an intravenous central catheter that will terminate in the superior vena cava, just above the right atrium. The primary benefit of central administration is that the vena cava is a large and robust vein that can accommodate more irritant medicines (such as potassium) than smaller vessels.
Peripheral administration	In reference to the delivery of a medicine by the intravenous route through an intravenous catheter that is generally inserted through the skin into a vein usually located in the hand, elbow, or foot (i.e., usually NOT a large, robust blood vessel)
Isotonic solution	Isotonicity refers to a solution that has the same osmolality, or solute concentration as another solution. In the context of this document, the designation of "isotonic" generally refers to a comparable osmolality to human plasma.
Extravasation	The leakage of a fluid out from its contained space (e.g., vein in this context) into the surrounding areas/tissues. The leakage of a vesicant fluid can contribute to possible tissue damage, irritation, and inflammation. The resulting impact will depend on the specific fluid involved, amount and degree of leakage and site/area affected.

Presentation

Concentrated intravenous potassium is readily available in two general formulations at SCGOPHCG:

1. **Standard Baxter® pre-mixed infusion bags** (containing varying concentrations of potassium)
2. **Concentrated ampoules/vials** (containing potassium)

Other formulations of intravenous potassium that are occasionally prescribed and used at SCGOPHCG are the non-standard potassium solutions.

These solutions are NOT proprietary products and can be described as customised potassium solutions that require preparation by a suitably qualified member of staff.

The use of a standard, proprietary potassium solution is the preferred and recommended practice in most situations as this minimises the likelihood for phlebitis and risk of adverse events.

Potassium content (in mmol)	Diluent	Volume	Isotonicity	Osmolarity (mOsmol/L)
Potassium Chloride 10mmol	sodium chloride 0.29%	100mL	isotonic	300
Potassium Chloride 20mmol	sodium chloride 0.9%	1000mL	hypertonic	348
Potassium Chloride 20mmol	glucose 5%	1000mL	mildly hypertonic	317
Potassium Chloride 40mmol	sodium chloride 0.9%	1000mL	hypertonic	388
Potassium Chloride 40mmol	glucose 3.96%	1000mL	isotonic	300
Potassium Dihydrogen Phosphate 10mmol	Sodium chloride 0.9%	250mL	hypertonic	365

** normal human reference range for plasma osmolality is approximately 275 to 299 milli-osmoles per kilogram
Printed or personally saved electronic copies of this policy are considered uncontrolled. Refer to the SCGOPHCG Policy Hub for current controlled electronic policies.

Table 1. Standard pre-mixed potassium infusion bags available from SCGOPHCG Pharmacy

Potassium Content (in mmol)	Volume
Potassium Chloride 10 mmol	10mL
Potassium Dihydrogen Phosphate 10 mmol	10mL
Potassium Acetate 25 mmol	5mL

Note: Solutions that involve the addition of potassium ampoules into a bag of sodium chloride 0.9% or glucose 5% are termed non-standard solutions.

Table 2. Concentrated potassium vials/ampoules available for use in preparing non-standard solutions only in critical care areas / approved clinical areas with access to cardiac monitoring facilities (except when approved by a consultant from the general ward area resourced to undertake appropriate monitoring)

For imprest availability, check Formulary One <https://formulary.hdwa.health.wa.gov.au/Imprest/Search>

Indication

For potassium replacement therapy in mild to severe hypokalaemia (< 3.5mol/L) only when oral supplementation is not possible or not clinically appropriate.

Dosage

Potassium can be supplemented via the enteral (e.g., oral, nasogastric, PEG) or intravenous route. Intravenous replacement of potassium should only be prescribed when there is no suitable alternative route following appropriate clinical assessment by a medical officer.

Prescribing Reminders

- Standardised prescribing of intravenous potassium is required.
 - Write out the full name e.g., potassium chloride.
 - Abbreviations such as *KCl*, *kayciel*, K^+ etc are not acceptable prescriptions and may lead to clinical errors.
 - Prescribe in millimoles rather than using 'milligrams per litre' or a 'percentage.'
 - Prescribe pre-mixed, standard potassium solutions where possible. Any non-standard potassium solution order should be confirmed verbally with the prescriber that the pre-mixed standard solutions are inappropriate and that a non-standard solution is required.
 - For more prescribing advice, refer to the [Practical Points section](#) below.

Preparation and Administration³

****CAUTION****

**Potassium should NEVER be administered subcutaneously or intramuscularly
Intravenous potassium chloride should NOT be prescribed as a rapid IV push or bolus
(10mmol/10mL) unless under exceptional circumstances (e.g. cardiac arrest) under the direction
of the most senior doctor present.**

IMPORTANT INFORMATION

- First-line option for intravenous replacement of potassium should always be standard pre-mixed solutions containing potassium chloride.

- Prescription of non-standard, more concentrated potassium solutions is permitted for patients in critical care areas with central venous access, **only following agreement with the consultant or senior medical staff on duty in the specific area.** *This involves the addition of concentrated potassium ampoules to solutions of either sodium chloride 0.9%, glucose 5% or other compatible fluid.*
- A rate-controlled pump must be used to infuse ALL potassium solutions to prevent unintentional bolus doses to be administered.
- Acceptable rates of administration of intravenous potassium differ between peripheral and central administration routes. Please follow the administration rates closely according to the dose of potassium prescribed and the intravenous access that your patient has available.
- Depending on the dose and rate of potassium administered, additional monitoring requirements may apply. See applicable section below for details.

The following section will be divided into two main parts:

1. Administration of **standard, pre-mixed potassium solutions**
2. Administration of **non-standard potassium solutions**

Please click  [hyperlinks](#) to the relevant part of this guideline.

Standard (pre-mixed) or Non-standard solution	Type of intravenous access	Hyperlink to administration details
Standard	Peripheral line	 click here
	Central line	 click here
Summarised administration table for standard formulations		 click here
Non-standard	Peripheral line	 click here
	Central line	 click here
Summarised administration table for non-standard formulations		 click here

STANDARD PRE-MIXED POTASSIUM SOLUTIONS ³⁻⁷

- Refers to the standard, pre-mixed potassium infusion bags available as a standard proprietary product.
- Pre-mixed potassium solutions will have a red/orange outer packaging.
- Strictly** no additional potassium or any other medication/electrolyte can be added to the pre-mixed solutions.
- Regular monitoring of patient serum potassium concentration and renal function should be undertaken.
 - Monitoring frequency is based on several factors such as haemodynamic stability, ongoing losses/underlying cause, ECG changes, renal function, other medications, etc, and should be made in consultation with a senior medical team member.
- Potassium dihydrogen phosphate may be used to treat [hypophosphataemia](#) (phosphate <0.8mmol/L). *Sodium dihydrogen phosphate is available if the patient requires IV supplementation and giving additional potassium would be inappropriate.*

Standard Potassium Solution Administration

1. Peripheral Administration

- Recommended rate of administration: 10mmol of potassium per hour. *
 - The **maximum rate of potassium administration** should **NOT exceed 10mmol per hour**.
 - Infusion rates >10mmol/hr are not approved at Osborne Park Hospital. Patients requiring these faster infusion rates must be transferred to SCGH.
- If giving potassium dihydrogen phosphate as a phosphate supplement for hypophosphataemia. [Please refer to the SCGH Hypophosphataemia Guideline.](#)

*** An exception exists in relation to the management of [DKA](#) and [HHS](#), where up to 20mmol per hour of potassium chloride may be administered peripherally with continuous cardiac monitoring.**

This exception has been reviewed and endorsed by the local SCGHPHCG MTC

It is strongly recommended to consider central access for patients in Critical Care Areas (ED, ICU) likely to require prolonged high doses (>20mmol per hour) of intravenous potassium supplementation.

2. Central Administration

- Recommended rate of administration: 10 to 20mmol of potassium per hour.
- The **maximum rate of potassium administration** should **NOT exceed 20mmol per hour** unless advised by a Critical Care Specialist or other Consultant.
- Continuous electrocardiogram (ECG) monitoring is required when the rate is faster than 10 mmol/hour.

Guide for Standard Pre-mixed Potassium Solutions

(refer to [PRESENTATION section](#) above for product details)

Potassium Administration	Peripheral line*	Central line
Recommended rate	10mmol per hour*	10mmol per hour
Maximum rate	10mmol per hour	20mmol per hour Rate in excess of this recommendation requires approval from a Senior Critical Care doctor
Continuous ECG monitoring	No	Yes; if rate is > 10mmol per hour

Note: Strictly NO additional potassium or any other medication/electrolyte can be added to these standard, pre-mixed solutions

A rate-controlled pump **MUST** be used to infuse ALL potassium solutions.

* In patients experiencing pain or discomfort upon administration, the rate may be extended to prolong the infusion time e.g., 10mmol over 2 to 3 hours.

Table 3. Recommended rate & requirements for administering standard pre-mixed potassium solutions.

NON-STANDARD POTASSIUM SOLUTIONS^{3,6-8}

(Involves the addition of potassium into a standard bag of sodium chloride 0.9%, glucose 5% or other compatible fluid)

- Prescription of **non-standard potassium chloride solutions** is permitted **only**:
 - if the patient is admitted to an intensive care or other critical care area where the supervising consultant/senior medical staff member responsible for the patient has authorised its use, OR
 - in exceptional circumstances where a standard, premixed potassium solution is considered unsuitable.
- The name of the registrar or consultant providing the approval must be documented on the 'Intravenous Fluid Order Chart' (i.e. 'approved by Dr.....').
- Potassium acetate is not available as a standard pre-mixed solution. If an infusion containing this formulation is required, then a non-standard solution will need to be prepared.

IMPORTANT INFORMATION

- Always consider the use of standard pre-mixed potassium solutions as first line option.
- Administration of concentrated, non-isotonic solutions of potassium (e.g. non-standard solutions) may cause pain and phlebitis.
- Following the addition of potassium chloride to the bag of compatible fluid, invert the solution at least 10 times to ensure adequate mixing and uniform distribution of the solute throughout.
 - Insufficient mixing by inversion results in uneven distribution and potential for adverse cardiac effects.
- Each bag of potassium should have two staff members check the final product, dose, dilution, labelling, route, and rate prior to administration.
- Strict policies exist relating to the concentration and rate of non-standard potassium solutions. This will depend on the type of venous access the patient has available (see below).
- Regular monitoring of patient serum potassium concentration and renal function should be undertaken.
 - Monitoring frequency is based on several factors such as haemodynamic stability, ongoing losses/underlying cause, ECG changes, renal function, other medications, etc, and should be made in consultation with a senior medical team member.

Non-standard Potassium Solution Administration**Peripheral Access Administration**

Note: First-line option should **ALWAYS** be the use of the standard pre-mixed potassium bags.

- The concentration of a non-isotonic non-standard solution of potassium for peripheral access must not exceed 40mmol/L (e.g., 10mmol per 250mL solution).
- Each 10mmol of potassium should be added to ≥ 250mL sodium chloride 0.9% or glucose 5%.
- More concentrated solutions may otherwise cause pain and phlebitis.
- If more concentrated solutions required,
 - Use the standard isotonic 10mmol/100mL potassium chloride solutions.
 - If a non-isotonic solution is deemed necessary, consider central access for administration.
- The **maximum rate** of potassium administration should **not exceed 10mmol per hour** except for acute management of [Diabetic Ketoacidosis](#) or [Hyperosmolar Hyperglycaemic State](#).
 - Patients must receive cardiac monitoring. Refer to the [Diabetic Ketoacidosis guideline](#) or the [Hyperosmolar Hyperglycaemic State guideline](#) for more detail.

- If giving potassium dihydrogen phosphate as a phosphate supplement for hypophosphataemia. [Please refer to the SCGH Hypophosphataemia Guideline](#).

Central Access Administration

10 to 20mmol (maximum) of potassium may be added to at least 100mL of sodium chloride 0.9% or glucose 5%.

The addition of 25mmol of potassium acetate requires a minimum of 125mL (if using a burette) or 250mL of solution.

- The **maximum rate** of potassium administration should **not exceed 20mmol per hour** unless advised by a critical care specialist / consultant.
- Continuous cardiac ECG monitoring is required for administration rates > 10mmol per hour.
- If giving potassium dihydrogen phosphate as a phosphate supplement for hypophosphataemia, the rate of infusion is slightly slower (due to risk of hypocalcaemia if phosphate is given too fast) (refer to Table 4 below).

Guide for Non-Standard Potassium Solutions (refers to the addition of concentrated potassium into a standard bag of compatible fluid)		
	Peripheral access*	Central access
Acceptable concentration of potassium per solution	10mmol per 250mL of diluent^ Consider the use of a standard, pre-mixed bag instead.	10 to 20mmol per 100mL of diluent^ If adding 10mmol only – use a standard, pre-mixed bag instead.
Maximum concentration of potassium per solution	10mmol per 250mL of diluent^ or 40mmol per 1000mL of diluent^	20mmol per 100mL of diluent^ Concentrations in excess of this recommendation require approval from a Critical Care Specialist.
Isotonicity of solutions	Hypertonic	Hypertonic
Recommended rate	10mmol per hour *	10mmol per hour
Maximum rate	10mmol per hour	20mmol per hour Rates in excess of this recommendation require approval from a critical care specialist.
Continuous ECG monitoring	No	Yes; if rate is > 10mmol per hour
Intravenous potassium chloride should NOT be prescribed as a rapid IV push or bolus (10mmol/10mL) unless under exceptional circumstances (e.g. cardiac arrest) under the direction of the consultant or most senior doctor present. - Always watch for potential extravasation risk, which may present as pain or phlebitis. - Avoid rapid administration of potassium as this may lead to potentially fatal hyperkalaemia. - Ensure careful monitoring of serum potassium concentration and administration rates prior to commencing each potassium infusion. In an emergency situation where the concentration exceeds 40mmol/L or rate is greater than 10mmol/hour, the infusion may be commenced through a peripheral line until central access is available. ^ sodium chloride 0.9% is the preferred diluent as glucose solutions may decrease potassium levels * In patients experiencing pain or discomfort upon administration, the rate should be extended to prolong the infusion time e.g., 10mmol over 2 to 3 hours. #amount of diluent and rate of administration may differ in the Intensive Care Unit (please refer to ICU guidelines)		

RESUSCITATION / EMERGENCY SITUATIONS⁹

As per the current Australian and New Zealand Committee on Resuscitation (ANZCOR) guidelines for Adult Advanced Life Support and specifically “Medications in adult cardiac arrest”:

- A bolus dose of 5mmol of potassium chloride may be administered intravenously in resuscitation situations such as persistent ventricular fibrillation due to documented or suspected hypokalaemia.
- Refer to the [ANZCOR Guideline 11.5 - Medications in Adult Cardiac Arrest \(Section 3.6\)](#)
- **NEVER** give potassium intramuscularly
- **NEVER** give potassium subcutaneously

NEAT (undiluted) POTASSIUM INFUSIONS

Neat, undiluted, concentrated potassium intravenous infusions are uncommonly prescribed at SCGOPHCG.

This refers to highly concentrated solutions of intravenous potassium chloride (**1mmol potassium chloride per mL**)

The use of highly concentrated potassium solutions is **not recommended** by national guidelines and may increase clinical risk from the perspective of unfamiliarity and inadvertent bolus dosing.

However, the use of neat, undiluted potassium infusions (50mmol potassium chloride in 50mL) may be deemed acceptable in the critical care setting and may therefore be prescribed following thorough clinical assessment and Senior Registrar / Consultant approval.

Concentrated potassium infusions are not recommended as routine options for potassium supplementation.

To accompany any prescription for a neat, undiluted potassium infusion:

1. The prescription must be prescribed by a Senior Registrar or Consultant.
2. Each infusion must be sighted and signed by two registered nurses prior to administration.
3. Hourly ABG monitoring for the patient is required during the duration in which the undiluted, concentrated potassium infusion is administered.
4. Maximum infusion rate of 20mmol of potassium per hour
5. Strictly NO bolus doses to be given.

Practical Points

Prescribing³

- Any prescription of intravenous potassium must be prescribed on an intravenous fluid orders chart (MR 410/825) or a SCGOPHCG ICU infusion chart (MR 410/825).
- The prescription of potassium salts should be written in its entirety and not abbreviated e.g. 'KCl,' 'Kayciel' or any other unacceptable abbreviation. Do NOT use chemical names or formulas such as KCl, KH₂PO₄ or C₂H₃KO₂ for potassium chloride, potassium dihydrogen phosphate and potassium acetate-. respectively⁹.
- Prescribing of potassium should follow the general principles for medication prescribing as per the hospital policies [HP246 SCGOPHCG Medicines Management Policy and Procedure](#) and [HP261 SCGOPHCG High Risk Medication Policy](#).
- Prescribing of intravenous potassium should be in millimoles rather than 'milligrams per litre,' 'milli-equivalents' or as a 'percentage.'
- All prescriptions for intravenous potassium solutions require a rate, route, dilution, and clear administration instructions.
- Rate of administration should be expressed as millilitres per hour (mL/hour).
- Prescriptions without these specified instructions should be considered incomplete and unacceptable.
- Do NOT use the term '*bolus*' for any IV potassium order unless under the exceptional circumstance of a cardiac arrest.
- Prescribe standard pre-mixed potassium infusions where possible.
- Neat, highly concentrated potassium infusions (e.g., 1mmol per mL) need to be approved and prescribed by a critical care consultant or senior registrar.

Supply³

Concentrated Potassium Ampoules

For General Ward Areas

- Concentrated potassium ampoules are NOT available as imprest items in general ward areas. This restriction complies with the recommendations outlined by the Australian Commission on Safety and Quality in Health Care (ACSQHC) and their national high risk medication alert in relation to intravenous potassium chloride.
- Their supply is restricted as per [MP 0131/20 High Risk Medication Policy](#).
- If there is an approved and appropriate prescription/order for the use of a potassium ampoule(s) by a consultant or senior medical staff member on a fluid order chart, then the order must be reviewed by a clinical pharmacist prior to arranging supply.
- The Department of Pharmacy will usually only supply the exact number of potassium ampoules required as per the prescription order.

For Critical Care Areas

- Concentrated potassium ampoules are available as imprest items ONLY in approved clinical areas with access to cardiac monitoring facilities.
- Potassium ampoules should NOT be transferred for use general ward areas, unless instructed by a clinical pharmacist or after-hours clinical nurse specialist (CNS).
- Following approval by clinical pharmacist or after-hours Clinical Nurse Specialist only the exact number of potassium ampoules prescribed will be provided to the general ward area.

Storage³**Standard Pre-mixed Potassium Solutions**

- Standard pre-mixed potassium solutions are stored as imprest items and available on all wards.
- They must be clearly labelled and separated from other similar sized intravenous solutions e.g., sodium chloride 0.9% 100mL bags.
- Store below 25°C.

Concentrated Potassium Ampoules**General Ward Areas**

- Concentrated potassium ampoules are NOT stored as imprest items on general ward areas.
- Concentrated potassium ampoules are available as imprest items ONLY in approved clinical areas with access to cardiac monitoring facilities.
- However, if an approved order for a specific number of ampoules for a designated patient has been ordered by the clinical pharmacist, then the ampoules should be stored in a clearly labelled bag from pharmacy and be assigned to the specific patient with their details.
- Until the ampoules are used, they may be stored in the patient's lockable medicine drawer, at temperature below 25°C and protected from light.
- Any unused ampoules MUST be returned to the pharmacy department or the clinical pharmacist on the ward immediately during pharmacy working hours. If the pharmacist is unavailable at the time (e.g., after-hours), store the unused potassium ampoules in a lockable drawer until the pharmacist is available.

Critical Care Areas

- Concentrated potassium ampoules may be stored as imprest items ONLY in approved clinical areas with access to cardiac monitoring facilities.
- Only one strength and size of potassium chloride ampoules will be stocked within the hospital
 - 10mL ampoules containing 10mmol potassium chloride.
- Potassium chloride ampoules should be stored and sealed in a clearly marked red container that is separated from other ampoules kept on imprest.
- Store at temperature below 25°C.
- Concentrated potassium ampoules must NOT be placed or stored on resuscitation trolleys.

Contra-indications^(4,5)

- When enteral (e.g., oral, nasogastric) administration of potassium is an acceptable option and clinically appropriate.
- In patients with hyperkalaemia ($K^+ > 5.2\text{mmol/L}$).

Precautions^(3,6)

- Confirm the dose and rate of potassium prescribed before administering each infusion as there are restrictions depending on the formulation (pre-mixed / concentrated bag), dose and choice of intravenous access that is used. A second staff member should also 'double-check' these requirements prior to administration.
- Rates $> 20\text{mmol/hour}$ are potentially hazardous and should only be permitted for patients admitted to a critical care area (e.g., ICU / CCU) with the agreement of the medical staff, with guidance of a department guideline /policy or area specific guideline.

- When non-standard potassium solutions are prepared from concentrated potassium ampoules the final solution must be inverted at least 10 times to ensure that the solute is thoroughly mixed throughout the solution.
- Potassium ampoules must never be added to a hanging infusion.
- Continuous ECG monitoring is required when the rate > 10mmol/hour and be under observation of a skilled staff member trained to accurately interpret an ECG.
- Concurrent administration of other medications via Y-site may change the infusion rate of potassium.
- Consider if administration of potassium infusion via a dedicated line can be facilitated.
- Caution when administering concurrent drugs via Y-site to ensure the potassium concentration and flow rates are not too high

Adverse Effects ⁽⁴⁻⁶⁾

- If pain at the infusion site occurs:
 - Reduce the rate of administration,
 - Use a pre-mixed isotonic solution.
- Concentrations > 30mmol per litre of potassium (which are non-isotonic solutions) may cause pain and phlebitis on administration.
- Extravasation may cause severe complications – watch for signs including pain at administration site.
- Potassium intoxication may present as:
 - Paraesthesia of the extremities, flaccid paralysis, listlessness, mental confusion, weakness and heaviness of the legs, hypotension, cardiac arrhythmias and heart block.
 - ECG abnormalities including disappearance of the P wave, widening and slurring of the QRS complex, changes in the ST segment and tall-peaked T waves. Nausea, vomiting, diarrhoea and abdominal discomfort.

Monitoring^(2, 4)

- Regular monitoring of patient serum potassium concentration and renal function should be undertaken.
 - Monitoring frequency is based on several factors such as haemodynamic stability, ongoing losses/underlying cause, ECG changes, renal function, other medications, etc, and should be made in consultation with a senior medical team member.
- Monitor for signs of extravasation when administering via a peripheral line:
 - pain, discomfort, burning sensation at infusion site.
- Continuous ECG monitoring is required when the rate of potassium administration is faster than 10mmol/hour.

Compatibility^(4, 6)

Standard pre-mixed solutions/bags - No additional medications may be added to these bags (including additional potassium salts)

- Caution when administering concurrent drugs via Y-site to ensure the potassium concentration and flow rates are not too high.
- **Compatible fluids** - glucose 5%, glucose 10%, glucose with sodium chloride solutions, compound sodium lactate, Ringer's, sodium chloride 0.45%, sodium chloride 0.9%.
- **Compatible drugs via Y-site** - aciclovir, adrenaline, amifostine, aminophylline, amiodarone, ampicillin, anidulafungin, atracurium, aztreonam, bivalirudin, calcium gluconate, caspofungin, cefazolin, cefotaxime, cefoxitin, cetaroline, ceftazidime, ceftolozane-tazobactam, ceftriaxone, ciclosporin, ciprofloxacin, cisatracurium, clindamycin, dexamethasone, dexmedetomidine, digoxin, dopamine, droperidol, edrophonium, ephedrine, eptifibatide, ertapenem, esmolol, fentanyl, filgrastim, fluconazole, furosemide, ganciclovir, gentamicin, glyceryl trinitrate, granisetron, heparin sodium, hydrocortisone, hyoscine hydrobromide, insulin (regular), isavuconazole, isoprenaline, labetalol, lidocaine, linezolid, magnesium sulphate, metaraminol, metoclopramide, micafungin, midazolam, milrinone, morphine sulfate, neostigmine, nicardipine,

noradrenaline, oxytocin, palonosetron, paracetamol, pethidine, piperacillin-tazobactam (EDTA free), posaconazole, ranitidine, remifentanyl, sodium bicarbonate, sodium nitroprusside*, suxamethonium*, tigecycline, tirofiban, tobramycin, vancomycin, verapamil, zidovudine

**Product information states incompatible however there is more recent compatibility information available demonstrating otherwise*

- **INCOMPATIBLE fluids** - fat emulsion, mannitol.
- **INCOMPATIBLE drugs** - amoxicillin, azithromycin, cefalotin, haloperidol, methylprednisolone, pentamidine, promethazine, thiopentone.

Related Legislation and Policy Documents

Key Related Documents:

- [Potassium Supplements \(Oral\) – SCGOPHCG Medication Management Guideline](#)
- [Hypokalaemia – SCGOPHCG Medication Management Guideline](#)
- [Adult Diabetic Ketoacidosis \(DKA\) - SCGOPHCG Medication Management Guideline](#)
- [Hypophosphataemia - SCGOPHCG Medication Management Guideline](#)
- [Adult Hyperosmolar Hyperglycaemic State \(HHS\) – SCGOPHCG Medication Management Guideline](#)
- [SCGOPHCG Nursing Practice Guideline No. 4 Intravenous Therapy](#)

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Policy:

- [MP 0131/20 WA DOH High Risk Medication Policy](#)
- [DOH Mandatory Standard for Intravenous Potassium 2020](#)
- [SCGOPHCG Hospital Policy 246 Medicines Management](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [SCGOPHCG Hospital Policy HP261 High Risk Medication](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)

Standards

- NSQHS Standard 1.1.1: An organisation-wide management system is in place for the development, implementation and regular review of policies, procedures and/or protocols.
- NSQHS Standard 1.7: Developing and/or applying clinical guidelines or pathways that supported by best available evidence (all subsequent actions of this criterion are applicable)
- NSQHS Standard 4.1: Developing and implementing governance arrangements and organisational policies, procedures and/or protocols for medication safety, which are consistent with national and jurisdictional legislative requirements, policies and guidelines.
- NSQHS Standard 4.1.2: Policies, procedures and/or protocols are in place that are consistent with legislative, national, jurisdictional and professional guidelines.

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10. Australian Commission on Safety and Quality in Health Care. [Recommendations for terminology, abbreviations and symbols used in medicines documentation](#). 2016.
11. DRG0043 [Potassium Drug Guideline – Intravenous infusion and enteral](#). Ballarat Health Service. Reviewed March 2023.

Authorisation & Version Control

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Standards Applicable:	Insert relevant NSQHS Standards (Version 2):  			

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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